

Lab 3: Implementation of DCL and TCL

1. DCL – Data Control Language

DCL is used to control access and permissions in a database. DCL mainly uses GRANT and REVOKE.

GRANT

GRANT is used to give privileges to a user or role.

Query 1 – Grant SELECT

GRANT SELECT

ON Student

TO Rahul;

This gives Rahul permission to retrieve data from the Student table.

Query 2 – Grant INSERT

GRANT INSERT

ON Student

TO Rahul;

Query 3 – Grant UPDATE

GRANT UPDATE

ON Student

TO Rahul;

Query 4 – Grant DELETE

GRANT DELETE

ON Student

TO Rahul;

Query 5 – Grant SELECT and INSERT

GRANT SELECT, INSERT

ON Student

TO Rahul;

Query 6 – Grant SELECT, INSERT and UPDATE

GRANT SELECT, INSERT, UPDATE

ON Student

TO Rahul;

The document notes that granting these privileges does not automatically give DELETE permission.

Query 7 – Revoke UPDATE

REVOKE removes a previously granted privilege.

REVOKE UPDATE

ON Student

FROM Rahul;

2. TCL – Transaction Control Language

TCL is used to manage transactions in a database. Its main commands include COMMIT, ROLLBACK, SAVEPOINT, ROLLBACK TO SAVEPOINT, and RELEASE SAVEPOINT.

For TCL examples, consider:

```
CREATE TABLE Accounts (  
    AccountID VARCHAR(10),  
    HolderName VARCHAR(50),  
    Balance INT  
);
```

Insert records:

```
INSERT INTO Accounts VALUES  
( 'A', 'Rahul', 50000),  
( 'B', 'Priya', 20000);
```

TCL Queries

1. BEGIN TRANSACTION

Starts a transaction.

BEGIN TRANSACTION;

Some DBMSs use:

START TRANSACTION;

2. COMMIT

COMMIT permanently saves the changes made during the transaction.

BEGIN TRANSACTION;

UPDATE Accounts

SET Balance = Balance - 10000

WHERE AccountID = 'A';

UPDATE Accounts

SET Balance = Balance + 10000

WHERE AccountID = 'B';

COMMIT;

After the transaction:

- Account A = ₹40,000
- Account B = ₹30,000

Memory: COMMIT = SAVE

3. ROLLBACK

ROLLBACK undoes changes made during the current transaction.

BEGIN TRANSACTION;

UPDATE Accounts

SET Balance = Balance - 5000

WHERE AccountID = 'A';

ROLLBACK;

The balance returns to its state before the transaction.

Memory: ROLLBACK = UNDO

4. SAVEPOINT

SAVEPOINT creates a checkpoint inside a transaction.

BEGIN TRANSACTION;

UPDATE Accounts

SET Balance = Balance - 5000

WHERE AccountID = 'A';

SAVEPOINT SP1;

Memory: SAVEPOINT = BOOKMARK

5. ROLLBACK TO SAVEPOINT

It undoes only the changes made **after a particular savepoint** while keeping the transaction active.

BEGIN TRANSACTION;

UPDATE Accounts

SET Balance = Balance - 5000

WHERE AccountID = 'A';

SAVEPOINT SP1;

UPDATE Accounts

SET Balance = Balance + 5000

WHERE AccountID = 'B';

ROLLBACK TO SAVEPOINT SP1;

COMMIT;

The changes after SP1 are undone, but the transaction remains active until COMMIT or another transaction-ending operation.

6. RELEASE SAVEPOINT

RELEASE SAVEPOINT removes a previously created savepoint.

BEGIN TRANSACTION;

UPDATE Accounts

SET Balance = Balance - 5000

WHERE AccountID = 'A';

SAVEPOINT SP1;

RELEASE SAVEPOINT SP1;

COMMIT;

Important: RELEASE SAVEPOINT **does not undo the changes and does not commit the transaction.** It only removes the savepoint.

7. Complete TCL Example

BEGIN TRANSACTION;

UPDATE Accounts

SET Balance = Balance - 10000

WHERE AccountID = 'A';

SAVEPOINT SP1;

UPDATE Accounts

SET Balance = Balance + 10000

WHERE AccountID = 'B';

ROLLBACK TO SAVEPOINT SP1;

COMMIT;

This demonstrates the complete flow:

BEGIN TRANSACTION

↓

Operations

↓

SAVEPOINT

↓

More Operations

↓

ROLLBACK TO SAVEPOINT

↓

COMMIT

↓

Save Changes

The document presents this same transaction flow using BEGIN, SAVEPOINT, ROLLBACK TO SAVEPOINT, and COMMIT.

Final Summary

Language Purpose		Commands
DQL	Retrieve data	SELECT
DCL	Control access/permissions	GRANT, REVOKE
TCL	Control transactions	BEGIN, COMMIT, ROLLBACK, SAVEPOINT, ROLLBACK TO SAVEPOINT, RELEASE SAVEPOINT

Easy Memory Trick

DQL → Query data

DCL → Control access

TCL → Control transactions

And for TCL:

COMMIT → SAVE

ROLLBACK → UNDO

SAVEPOINT → BOOKMARK

ROLLBACK TO SAVEPOINT → PARTIAL UNDO

RELEASE SAVEPOINT → REMOVE BOOKMARK